MSFS_AutoFPS v0.5.5.1

Notice

My future development efforts on this app are mainly limited to maintenance, resilience improvements and streamlining of existing functionality only. I do add new functionality at times, mainly from my existing wishlist. I occasionally accept user requests for new functionality, however these will only be accepted if I judge it to be a great idea and it is technically achievable, useful to the majority of users, consistent with AutoFPS's existing design philosophy, with negligible, or preferably no, UI impact, and if I have the available time to do it.

Licence

- Versions released before 8 November 2026, up to and including v0.5.5.0, were released under the MIT Licence (with an exception for the app icon image and associated files, and a requirement to attribute previous authors) and remain under it.
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- The only official source of this app is github.com/ResetXPDR/MSFS_AutoFPS. Copies, forks or rebuilds published elsewhere, including any rebuilt from decompiled binaries, are not made, endorsed or supported by me. Please download only from the official Releases page.

Important Simulator Notes

- **Official Forums Warning:** Do not mention or discuss this app on the official MSFS forums, even in private messages. Forum moderation treats active memory access as a Terms of Service violation regardless of its harmless operation. If you post about it there, your account may be flagged or banned.
- **SimConnect Wishlist:** There is an official MSFS wishlist item requesting SimConnect variable access to graphics settings, which would allow legitimate settings

management. Please vote for it here.

- **Memory Access & Safety:** AutoFPS reads and adjusts active simulator graphics settings in memory at a maximum rate of once per second per setting. On startup, memory offsets are validated; if an unsupported simulator build is detected and updated offsets cannot be validated, the app restricts itself to read-only mode to prevent unintended changes. AutoFPS is provided as-is and used at your own risk.

Table of Contents

This readme goes from the basics to the full detail. Sections 1 to 3 get you installed and running, sections 4 to 7 explain how to use and read the app, sections 8 and 9 cover Expert mode and the companion apps, section 10 helps when something goes wrong, and section 11 holds the complete technical detail.

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Summary

Based on muumimorko's idea and code in MSFS_AdaptiveLOD, as further developed by Fragtality in DynamicLOD, myself in DynamicLOD_ResetEdition and MSFS2020_AutoFPS and with inspiration from changes suggested by kayjay1c6b from his MSFS2024_AutoFPS and by dankocrnkovic from his fork of MSFS_AutoFPS.

Now fully compatible with MSFS 2020 and 2024 in a single application, AutoFPS improves simulator smoothness and framerate consistency by automatically changing key graphics settings on the fly. Key features include:

- Automatic TLOD adjustment to achieve and maintain a target FPS or follow an altitude schedule.
- Advanced FPS smoothing (FPS+) to filter out brief rendering spikes and dips.
- In-Game Toolbar Widget and local web interface (`http://localhost:54321`) with real-time telemetry, camera-based auto-hiding, and cockpit controls.
- Dynamic graphics mode architecture supporting native rendering, VR, native Frame Generation, and external mods (MSM, VectorXR, OFXR, Lossless Scaling) with dual-layer VR conflict protection.
- Simplified Non-Expert mode with automated guardrails suited to most user setups.
- Deep Expert Mode customisation covering sensitivity models, altitude bands, night optimisations with configurable day/night switching, multi-profile pairing (VR, sim rate and Day/Night), and CPU thread optimisation.
- A **Monitor only** flight type that stops AutoFPS changing any simulator setting and simply monitors performance.
- Native Dark Mode theme, resizable fluid layout, and automatic restoration of simulator settings upon exit, enhanced to withstand MSFS CTDs.
- Auto future MSFS version compatibility, provided MSFS memory changes are minor, plus detection of and protection from known similar apps already running or incompatibilities with newer MSFS versions.
- Intelligent auto-update system that verifies each downloaded installer (SHA-256), defers widget updates until MSFS exits, and supports custom Community folder and Addons Linker management that persists across updates.
- Optional MSFS Performance Optimiser, which selects the best CPU core affinity, process priority and power plan for MSFS.

Installation & Updates

Basically: run the installer to install, update or uninstall the app. If anything goes wrong, see Troubleshooting & FAQ.

Requirements

- Microsoft Flight Simulator 2020 or 2024
- Windows 10 / 11 (64-bit)
- .NET 8 Desktop Runtime (x64)
- Visual C++ 2015–2022 Redistributable (x64)
- MobiFlight WASM Module (managed by the installer)

Installing

1. Download the latest installer (`MSFS_AutoFPS-installer-vX.X.X.X.exe` , about 3 MB) from GitHub Releases.
2. Make sure MSFS_AutoFPS and/or MSFS2020_AutoFPS are not running. If the MobiFlight module is missing or outdated, MSFS must be closed too.
3. Run the installer, choose your options and install. Do not run it as Admin unless it will not install because of a permissions issue.
4. Start AutoFPS, ideally just after MSFS has loaded to the main menu, then follow the Quick Start.

Installer Options

- **Desktop Icon:** Creates a desktop icon for the app.
- **MSFS Widget:** Deploys the AutoFPS In-Game Toolbar Widget to the Community folder of any installed MSFS versions.
- **Close to Tray:** The app closes to the system tray.
- **Reset Configs:** Resets the app Common, MSFS 2020 and MSFS 2024 config files. When upgrading from 0.5.5.0 or earlier, the config files are first backed up to a `ConfigBackup_<version>` folder.
- **Repair:** Reinstalls core files, including the latest Visual C++ Redistributable and .NET 8 runtime, MobiFlight and the In-Game Widget (if enabled), keeping your app config intact.
- **Auto Start Options:** Remove, Retain, FSUIPC (via `FSUIPC.ini`) or MSFS (via `EXE.xml`).
- **Custom Community Folders:** Use `2020 Folder...` or `2024 Folder...` to point to custom Community paths (such as Addons Linker locations). Custom paths persist across updates in `Install.log` . Right-click either button to reset to the auto-detected simulator defaults.

Updating, Upgrading and Uninstalling

- **Updating:** Just run the installer and select update. Do **not** uninstall first, because that deletes all app files including your config.
- **Upgrading from MSFS2020_AutoFPS:** No need to uninstall it first. The installer removes it and preserves its MSFS 2020 config for use in this app.
- **Uninstalling:** Exit the app completely (not hiding in the system tray), run the installer and select **Remove** on the first window. This removes the desktop icon, any MSFS or FSUIPC autostart entries, and the entire app folder including your config file and the in-game widget.

Update Modes & Channels

- **Silent Updates:** Installs updates automatically without prompts, displaying release notes in Notepad afterwards.
- **Prompted Updates (Default):** Prompts for confirmation before installing optional updates.
- **Show Updates:** Displays notifications when updates are available without downloading.
- **Mandatory Updates Only:** Notifies and installs only mandatory builds.
- **Update Channels (Expert mode):**
  - **Release:** Standard stable release baseline.
  - **RC+ (Release Candidate), Test+, Exp+ (Experimental):** Accesses pre-release builds. Returning from a test channel to Release offers a one-time automatic rollback to the latest formal release. Test versions require all app update notifications to remain active to maintain a consistent test baseline.

Safe Updates

- **Verified:** Before launching a downloaded installer, auto-update checks its SHA-256 against the digest GitHub publishes for the release. On a mismatch it downloads the installer again, and if no digest is available the update is refused.
- **Deferred:** When an auto-update contains an in-game widget update while MSFS is running, the installer defers installation until MSFS closes, preventing flight interruptions and locked-file errors.

File Locations

The installation location is fixed to `%appdata%\MSFS_AutoFPS`.

- Binaries: `%appdata%\MSFS_AutoFPS\bin`
- Logs: `%appdata%\MSFS_AutoFPS\log`
- Installation Log: `%appdata%\MSFS_AutoFPS\Install.log`
- Configs: `%appdata%\MSFS_AutoFPS\MSFS_AutoFPS.config` (common), `MSFS2020_AutoFPS.config` and `MSFS2024_AutoFPS.config`

Quick Start & Non-Expert Guide

If you are new to the application or prefer an automated set-and-forget experience, Non-Expert mode provides reliable performance management without complex tuning.

Quick Start Steps

1. Configure your general graphics settings inside MSFS as you normally would.
2. Launch AutoFPS before loading into a flight.
3. Ensure **Expert** (Use Expert Options) is left unchecked.
4. Select your **Flight Type**:
    - **VFR:** Tuned for lower altitudes, GA aircraft, and higher visual expectations.
    - **IFR:** Tuned for airliners, higher climb rates, and complex airport performance.
    - **Monitor only:** Choose this to watch performance without AutoFPS changing any settings (see Monitor Only).
5. Set your **Target FPS**:
    - **Auto:** Recommended if you want the app to establish a realistic target based on your system headroom.
    - **Manual:** Set to a framerate your hardware can comfortably sustain in moderate conditions.
    - **Fixed:** Set to match your lock if using a driver or monitor refresh lock.
6. Click back into MSFS, allow any initial FPS settling or seeking events to complete (up to 60 seconds), and fly.

Non-Expert Automated Guardrails

In Non-Expert mode, AutoFPS manages LOD and visual settings using conservative, pre-tuned rules to prevent system overloads:

- **Automation Method:** Uses FPS Sensitivity at 5% for smooth, progressive adjustments.
- **Base Min TLOD:** 50% of your simulator setting for VFR, 25% for IFR (doubled if Auto Target FPS is selected), clamped to a maximum ceiling of 150.
- **Base TLOD Altitude:** 100 ft AGL for VFR, 1,000 ft AGL for IFR.
- **Top Max TLOD:** 250% of baseline for VFR (capped at 400), 200% for IFR (capped at 400).
- **Ground OLOD:** Capped at 200 (VFR runs at 150% of baseline up to 200; IFR runs at 100%).
- **Night Optimisation:** Automatically uses `TLOD÷2` at night (capped at 200) to shed distant scenery workload in the dark. By default, night starts at about the end of civil twilight.
- **MSFS 2020 Specifics:** Decreases cloud quality if performance cannot be recovered at Base Min TLOD or if GPU load exceeds 98% (when GPU-Z is running).
- **MSFS 2024 Specifics:** Engages automated settings reduction under marginal conditions and supports VRAM+ overflow mitigation when GPU-Z is active.

Note: Do not manually change graphics settings in the MSFS menu while in an active flight with AutoFPS running, as the app will overwrite them or restore original values when the session ends.

Desktop App Window & Controls

- **General Operation:**
  - Can be started anytime, but preferably just after MSFS loads to the main menu. The app exits automatically when MSFS closes.
  - Set `ExitAppAfterFlightSession = true` in `MSFS_AutoFPS.config` if you want the app to exit automatically at the end of each flight.
  - Running as Administrator is not typically required, but AutoFPS must run at the same elevation level as MSFS. When running elevated, `Admin` appears in the title bar.
- **Window Positioning & Memory:**
  - Window position and minimised/maximised state are remembered across sessions (except movements made in VR).
  - Disable position saving by setting `RememberWindowPos = false` in the configuration file.
  - The window automatically resets to its default position `(50, 50)` if restarted within 15 seconds of closing. Disable this by setting `AllowWindowPosReset = false`.
- **Title Bar Hiding & Whole-Window Dragging:**
  - The title bar can be hidden or restored by double-clicking the title bar, or via the right-click context menu (**Hide Title Bar**).
  - Left-click and drag anywhere on empty body or panel space to move the window.
- **Progressive UI Collapsing:**
  - Double-clicking the title bar area progressively collapses the desktop interface:
    - First double-click: Hides the Expert settings section (if active).
    - Second double-click: Hides the General settings section.
    - Third double-click: Restores all hidden sections.
- **Right-Click Context Menu:**
  - Right-clicking anywhere on the window provides instant access to **Hide/Show Title Bar**, **Minimise**, **Close to Tray** (if selected during installation), and **Exit** (exits immediately without prompts).
- **Streamlined Close Confirmation:**
  - The confirmation prompt only triggers when clicking the window close `[X]` button. Deliberate multi-step exits (`Alt + F4` , right-clicking the Taskbar icon and

selecting *Close window*, or using the context menu) exit immediately.

- **Main Controls:**
  - **Use Expert Options (Expert):** Unchecked (default) uses the built-in guardrails. Checked reveals the additional settings. If you do not understand these settings and their impact on MSFS performance and graphics quality, it is strongly recommended that you leave this unchecked.
  - **Flight Type:** VFR or IFR in any mode, plus six additional user profiles in Expert mode, and **Monitor only** in any mode, which stops AutoFPS changing any sim settings (see Monitor Only).
  - **Target FPS:** The most important setting in the app (range 10–500, fractional values of 0.5 supported). Set it to the FPS you want after any frame generation has been applied.
  - **Target FPS option (Manual, Auto, Fixed):** How the Target FPS is applied (see Graphics Modes & Frame Generation). The choice is saved separately for each graphics mode in each profile.
  - **On Top:** Lets the app overlay your MSFS session, with MSFS keeping the focus. Useful for adjusting settings and watching the result as the flight progresses, and for single monitor users with frame generation, who then see the true FG FPS the app is reading.
  - **Dark Mode:** Toggles between the default Light theme and a native Dark theme.
  - **Reset button:** Resets TLOD, Clouds, Auto Target FPS and graphics mode detection to their initial state, which is useful to restart the TLOD Extra seek if conditions change a lot. In Expert mode, outside a flight session, it restores the active flight profile to defaults after confirmation. Press **ALT-R** while the app has the focus, so it can be assigned as a VR-friendly voice command with an app like VoiceAttack.
  - **20>24 / 24>20 button:** If the Sim Version shows in red and is not the MSFS version you want to configure before starting it, click this button to switch to that version.

Reading the Display: Colours and Symbols

Connection Status

- **Red** means not connected and **green** means connected. The Sim Version shows **royal blue** when the MSFS Performance Optimiser is enabled and controlling MSFS.
- The **Widget** label (shown if the In-Game Widget is installed) is green when the toolbar widget or the browser page (`http://localhost:54321/`) is open and receiving telemetry, and red when closed or inactive.

- Sim Values only show valid values when all three connections are green (or royal blue for the Sim Version with the Optimiser). `n/a` means not available right now.

Sim Values colours

- **Green:** the value meets or exceeds the target.
- **Red:** the value is at its minimum or below target.
- **Orange with arrows (TLOD or OLOD):** the value is auto-adjusting.
- **Orange (FPS and TLOD):** adjustment is limited because FG is inactive and TLOD increases are disabled.
- **Black:** none of the above.
- Other symbols: 🔒 value locked, ▲ increasing, ▼ decreasing, `T` upper limit reached (also shown beside TLOD while VRAM+ is limiting it, with an explanatory tooltip), ⊥ lower limit reached.
- The Sim Values header shows an **MSFS icon** when default MSFS values are being displayed (not in a flight session), and an **AutoFPS icon** when the values are being actively controlled by AutoFPS in a flight. Both icons show when hybrid dynamic settings is active.
- Mouse over the **Reduce** value to see the additional reduction settings when auto reduction or VRAM+ is at Level 1 or greater.
- The **Reduce** label shows `DynSet XX` or `Inactive` for only 15 seconds after a flight starts, a full reset, or MSFS dynamic settings being switched on or off (Expert), then returns to the normal cloud state unless settings reduction is active.

FPS display

- **FPS** shows the unfiltered raw FPS when a Fixed FPS is set or detected. **FPS+** shows the average FPS, filtered for spikes and dips, so brief changes (view changes, panning, scenery loading) do not cause unwanted setting changes.
- The FPS value shows in **two alternating purple shades** while detailed FPS logging is running (Log+ mode). Click it again to cancel.
- The FPS source icon shows RTSS, MSFS, OFXR or VXR. Click it to switch between MSFS and RTSS when RTSS is running.

Target FPS dropdown

- The dropdown background turns **orange** when the Target FPS type selected is not the one for the current graphics mode, and also when Target FPS is higher than a detected FPS cap, with a tooltip explaining how to fix it.

Status line

- The sim rate shows at the start of the line for any value except 1X.
- Shows the detected graphics mode, app pause, FPS settle, Mtns+, app priority and the TLOD range as applicable. A + next to TLOD means TLOD Extra is active, so expect a higher TLOD Base Min.
- **Day**, **Night** or **Δ** shows the day/night state, with Δ while transitioning between day and night. The same state selects the Night profile in a Day/Night profile pair.
- A 🔒 next to the TLOD range means the delayed base TLOD reduction (`DelayLG`) is active and waiting for the landing gear.
- A **lightning symbol** next to the FPS reading means periodic frame time spikes have been detected, and next to the TLOD range means spike protection is currently reducing TLOD.
- When VRAM+ is reducing settings, the **Reduce** label changes to **VRAM+** in red, and the status line appends **HLD** (hold) or **RED** (reduction).
- A **D** prefix on the TLOD and OLOD labels means MSFS dynamic settings is active and controls the LOD reductions.

In-Game Toolbar Widget & Web Interface

The companion In-Game Toolbar Widget and local web server allow in-cockpit monitoring and adjustments without switching out of the simulator.

Installation & Access

- Select the **MSFS Widget** option during the installer setup to deploy the widget to your Community folder.
- In-cockpit: Open the MSFS top toolbar and click the **AutoFPS** icon.
- Web browser: Navigate to `http://localhost:54321/` in a browser on the PC running AutoFPS while it is running. The web server only accepts requests addressed to localhost / 127.0.0.1.

Key Widget Features

- **Live Telemetry & Balanced Layout:** Mirrors the desktop application across three clean columns displaying real-time FPS, TLOD, OLOD, AGL, vertical speed, reduction states, and connection status with matching glyphs.
- **Interactive Target FPS Controls:** Adjust Target FPS directly in the cockpit using responsive `–` and `+` stepper buttons (with accelerating hold), toggle modes (Manual, Auto, Fixed), and switch target graphics profiles.

- **Camera-Based Auto-Hide:** Automatically conceals the in-game toolbar panel when leaving cockpit view to keep external views and screenshots uncluttered:
  - **Off:** Widget remains visible in all camera views.
  - **Chase:** Automatically hides only in standard chase exterior view.
  - **All Ext:** Automatically hides across all external views (chase, drone, showcase, and exterior fixed cameras) with sub-150 ms SimVar response.
- **Click-to-Toggle FPS Source:** Click the FPS source badge on the panel to toggle capture between MSFS and RTSS (read-only during active OFXR or VectorXR sessions).
- **Show/Hide Hotkey:** The previous on-screen hotkey button has been replaced by the Auto-Hide dropdown. Any previously configured hotkeys continue to work in the background. To set, change, or clear a manual toggle hotkey, edit the `WidgetHotkey` key in `MSFS2020_AutoFPS.config` or `MSFS2024_AutoFPS.config` located in `%appdata%\MSFS_AutoFPS` (set to `"None"` to disable).
- **Collapsible General Controls:** Click the General section header to collapse controls during flight, synchronised two-way with the desktop application.
- **Title Bar Hiding & Dragging:** Double-click empty space on the widget header to toggle the title bar. Move the widget by dragging anywhere on its background. Text and elements dynamically scale when resized horizontally to prevent clipping in VR.
- **Monitor Only Layout:** When the Flight Type is **Monitor only**, the widget (and web interface) hides Target FPS, Expert and Reset, moves Dark to the Flight Type line, and shows Log+ locked on and dimmed. Target FPS, mode, Target, Expert and Log+ changes made from the widget are ignored while in Monitor only.
- **Flight Type List:** The list is rebuilt each time it is opened with the selected item scrolled into view, scrolls with the mouse wheel and has a wide scrollbar that can also be clicked or dragged.

Graphics Modes & Frame Generation

AutoFPS incorporates a dynamic graphics mode architecture. Standard native modes are permanently available, while external frame generation mods populate automatically only when their background processes or API layers are detected.

Target FPS Guidelines (Fixed vs NTE vs Manual)

- **Fixed FPS (True Lock):**
  - Use when your framerate is rigidly locked to an exact number (e.g. monitor refresh divisor, driver limiter, or Reflex lock).
  - Select **Fixed Target FPS** and match your lock.

    - Target FPS can be set between 0% and 5% below a detected cap without warning to accommodate standard G-Sync, FreeSync, and Reflex frame-rate limits. If set more than 5% below the cap, the app advises switching to Manual mode or raising the target to match.
- **NTE FPS (Not-To-Exceed Cap):**
    - Use when a ceiling cap exists but framerate still fluctuates below it.
    - Select **Manual Target FPS**.
    - Target FPS must be set at least 5% below the cap, ensuring settings reduction has sufficient headroom to recover.
- **Auto Target FPS:**
    - Use when you don't know what target to choose, or your flights are so varied that one target is not always right.
    - It calculates a target after the initial FPS settling, when stationary on the ground or any time you are in the air, and recalculates if performance becomes too low, after arriving at a new destination, when changing graphics mode, pressing Reset, or toggling the option off and on.
    - It detects a fixed FPS (stable with minimal variation) and, when found, sets that as the target and switches to Fixed FPS automation.
    - Otherwise, with IFR or an Expert profile, the target ranges from 95% of the current average FPS on the ground to 85% at or above 3000 ft, leaving headroom for Max TLOD. VFR reduces each percentage by 5%.
    - Auto settings reduction, TLOD Base Extra and Free are mutually exclusive with Auto Target FPS, so Fixed TLOD must be used.
- **Warning System:** Setting Target FPS strictly higher than a detected cap triggers an orange mismatch highlight with corrective tooltips (buffered by 0.5 FPS to avoid timing jitter alarms).

Supported Graphics Modes

- **NFR (Native Frame Rate):** Standard 2D rendering without frame generation.
- **VR:** Native OpenXR virtual reality rendering inside MSFS.
- **nVidia FG (DLSS 3 / FG Mod / MFG):** Native frame generation. Requires MSFS window focus; disengages when the simulator loses focus.
- **FSR FG:** AMD frame generation in MSFS 2024 (2× multiplier), operating regardless of window focus.
- **MSM (MFG Super Motion):** Autodetected when the mod process is running. Parses dynamic multipliers (2X–6X, 1X Adaptive) with independent IFR/VFR profiles and tracks window focus (`MSM Inact` at base framerate when unfocused).

- **Lossless Scaling FG (LSFG):** Autodetected when Lossless Scaling is running. Adaptive mode operates as LSFG using the pre-LSFG multiplier.
- **OFXR (Optical Flow OpenXR):** Dedicated VR graphics mode for the OFXR Bridge Tray app across community builds. Features sub-50 ms controller loop synchronisation, auto-derived multipliers, and pacing desync settle recovery.
- **VXR (VectorXR):** Dedicated VR mode integrating directly with VectorXR via Status API v1. Reads live compositor telemetry, reports fractional multipliers (e.g. `VXRDyn 2.4X`), and handles standby states.
- **Dual-Layer VR Conflict Protection:** If both OFXR and VectorXR OpenXR layers are detected executing concurrently, AutoFPS warns the user, reverts to standard VR mode (`VR Conflict!`), and pauses LOD automation to avert frame pacing contention and VRAM spikes.
- **Man FG (Manual Frame Generation):** Forces manual multipliers (2X to 8X or DynFG) for unsupported or external frame generation utilities.

Using the Graphics Modes

- Core modes (NFR, VR, FG, FSR, Man FG) are always present. External mods (OFXR, VXR, LSFG, MSM) only appear when detected, and AutoFPS falls back to NFR if they close.
- The app starts in the last-used mode and switches to the detected mode when a flight loads. **Man FG** is a forced mode: it stays selected across restarts until you choose Auto to return to detection.
- Each graphics mode and flight type has its own Target FPS and its own Manual, Auto or Fixed option. You can edit an inactive mode's target and option in the dropdowns, which is handy for VR because you can set it in 2D before putting the headset on.
- Dynamic or adaptive FG set outside MSFS should be configured under **Man FG** (DynFG). With MSFS as the FPS source, Target FPS is the NFR (1×) value. With RTSS as the FPS source, Target FPS is a post-FG value, the active multiplier is shown in the status line, and TLOD automation is disabled because pre-FG frametimes cannot be derived reliably.
- **nVidia FG** only works while MSFS has the focus (driver behaviour), so use the app's **On Top** option to keep MSFS focused.
- **Lossless Scaling** needs v2.13.2 or later, is detected whenever LS is running, and uses a profile named `MSFS2020` or `MSFS2024`, falling back to the first profile. Press Reset after changing LS settings mid-flight, and use only one FG type at a time.
- See the Detailed Reference at the end for the full behaviour of each mode.

Expert Mode Reference Guide

Enable **Expert** (Use Expert Options) to reveal deep customisation controls across automation algorithms, altitude transitions, and scenery offsets.

TLOD Automation Methods

- **FPS Sensitivity (Default & Recommended):** Adjusts TLOD in smaller, more frequent increments proportional to the variance between current and target FPS. Lower sensitivity values (e.g. 2–3) produce smaller steps, ideal for resolving visual stutters. Step size equals double the sensitivity value.
- **Auto TLOD:** Disregards FPS targets and automates TLOD strictly according to aircraft altitude (AGL), interpolating between Base Min and Top Max.
- **Retired methods:** FPS Tolerance and FPS Cap were retired in 0.5.5.1. Profiles using them are converted automatically, with a message: FPS Tolerance profiles become FPS Sensitivity, and FPS Cap profiles become FPS Sensitivity with Fixed Target FPS for every graphics mode. Use FPS Sensitivity with Fixed Target FPS for locked framerates.

TLOD Altitude & Range Settings

- **TLOD Base Min:** Minimum TLOD applied at or below Alt TLOD Base (Range: 10 to Top Max - 10).
- **Alt TLOD Base:** Altitude (AGL) at or below which TLOD remains at Base Min (Range: 100 ft to 100,000 ft).
- **TLOD Top Max / Min:** Ceiling TLOD applied at or above Alt TLOD Top (Range: Base Min + 10 to 1,000).
- **Alt TLOD Top:** Altitude (AGL) at or above which TLOD can reach Top Max. Tooltips calculate recommended climb/descent rates (defaulting to 1,350 fpm for VFR and 2,000 fpm for IFR).
- **TLOD Base Extra:**
  - **Fixed:** Disables extra ground seeking.
  - **Extra:** Progressively seeks higher base TLOD when parked or taxiing (≤ 5 knots) under favourable headroom, freezing or reducing during takeoff rolls and low flight.
  - **Free:** Allows free-ranging TLOD between Base Min and Top Max at all altitudes based on real-time headroom (recommended only for rural VFR).
  - **Start Max:** Begins flight sessions with maximum extra TLOD rather than ramping up from base. It is skipped when VRAM is already above the VRAM+ recovery

threshold at session start, and cancelled if VRAM reaches the VRAM+ reduce threshold during the FPS settle.

- **Taxiing and Takeoff:** TLOD changes are allowed while taxiing at up to 20 knots, but not during a takeoff roll (above 20 knots, or above 5 knots while accelerating). The same rule gates cloud quality increases and settings reduction recovery.
- **Mountains+ (Mtns):** Expands Top Max TLOD by a set offset when terrain elevation exceeds a defined threshold (e.g. 1,000 ft AGL).
- **Night Mode (Per-Profile):** A dropdown in each profile:
  - **Off:** Maintains standard daytime settings around the clock.
  - **TLOD÷2:** Halves Top Max TLOD at night to eliminate distant rendering that cannot be seen in the dark.
  - **Clds-X:** Steps down volumetric clouds by the Night cloud reduction levels (never below Low), keeping full draw distance for night city lights and runways while reducing cloud load.
  - **Both:** Applies both TLOD halving and cloud reduction simultaneously.
  - *Night cloud reduction levels:* One global setting per sim config file (`NightCloudReduceLevelsGlobal` , 1 to 3, default 2) used by all profiles, Non-Expert included. VR has its own value (`NightCloudReduceLevelsGlobalVR` , 1 to 3, default 1), because VR cloud quality is usually lower than 2D on the same system. The Clds-X label shows the value for the graphics mode selected in Target. Night cloud reduction and its daytime restore apply straight away, not after the FPS settle.
  - *Switching:* Day or night is detected from the sim's automatic brightness, so timing follows the actual light level including latitude, season and cloud cover. `DayNightSwitchLevel` (0 to 10, default 5) in `MSFS2020_AutoFPS.config` / `MSFS2024_AutoFPS.config` moves the switch earlier or later: 0 is about sunrise / sunset, 5 is about the end of civil twilight and 10 is near total darkness. Set it to 0 for the pre-0.5.5.1 timing.
  - *Hold:* After each flip, further flips are held for 5 minutes of sim time (`DayNightSwitchHoldMinutes` , 1 to 30), so flying in and out of broken cloud does not flip night settings on and off. The hold ends early if brightness moves `DayNightSwitchHoldLevels` levels (0 to 2, default 2) past the switch level. A manual sim time change ends any hold and day/night is not decided until 3 seconds after the last change.
  - *Behaviour:* Automatically clears any active daytime cloud boost (`IncCloudQ`) upon entering night mode and restores it at sunrise. Active (or disabled) settings reduction takes priority over night cloud reduction in the Clouds / Reduce display, with the night reduction shown in the tooltip.

Auto OLOD

- Adjusts Objects LOD based on an altitude band independent of FPS to avoid automation contention.
- Interpolates between **OLOD @ Base** (at or below Alt OLOD Base) and **OLOD @ Top** (at or above Alt OLOD Top). Allows higher OLOD on the ground for airport detail and lower OLOD at cruise, or vice versa.

Flight Profiles & Command Line Arguments

- Supports up to 12 fully customisable profiles. Adjust the available total by editing `ProfileCount` (range 4–12, default 8: VFR, IFR, User 1–6) in `MSFS_AutoFPS.config`.
- **Profile Renaming:** Double-click the profile dropdown to rename. Press Enter or click away to save.
- **VR Auto-Switching:** Append `" VR"` to a profile name matching a 2D profile (e.g. `IFR` and `IFR VR`). AutoFPS switches profiles automatically when entering or exiting VR (⇔ symbol displayed).
- **Sim Rate Switching:** Append `" xX"` (e.g. `IFR 1X` and `IFR 4X`) to trigger automatic profile switching when acceleration rates change.
- **Day/Night Pairing:** Name two profiles the same, with one ending in `" Night"` (e.g. `IFR` and `IFR Night`), and AutoFPS uses the Night profile at night and the other during the day (Day or Night shows on the status line). See the Detailed Reference for how the two profiles interact.
- **Name Matching:** VR, Night and sim rate pairing ignores capitalisation and unusual spacing (e.g. `IFR night` or `IFR  2x` still pair). Profile names are cleaned on load and when renamed.
- **Landing Gear Delay (`DelayLG`):** Include `"DelayLG"` in a profile name to lock TLOD at Top Max during descent until the landing gear is lowered.
- **Command Line Switches:** Launch the application with predefined profiles:
  - `-ifr` and `-vfr`: Launches AutoFPS with the default IFR or VFR profile (even if renamed).
  - `-user1` through `-user6`: Launches directly into the corresponding User profile (Expert mode).
  - `-profile "<profile name>"`: Loads the exact specified profile name (Expert mode).

MSFS Performance Optimiser

Enabled via the `+` checkbox beside the Sim Version label. Directs thread affinity and scheduling to improve simulator frame pacing:

- **CPU Affinity:** Enforces physical-core-only execution (excluding SMT/Hyperthreads). On Intel hybrid architectures, restricts MSFS to performance cores (P-cores). Dual-CCD

AMD processors can enable first-CCD affinity.

- **Process Priority:** Automatically elevates MSFS process priority from Normal to High.
- **Power Plan:** Overrides Windows Balanced plan to High Performance or Ultimate Performance when the sim is active.
- *Safety:* Stands down if external utilities (Process Lasso, VR Auto Optimiser) have already adjusted simulator affinities.
- **Config File Parameters (`MSFS_AutoFPS.config`):**
  - `AffinityPhysicalCoreThreshold` (default `6`): SMT-core threshold at which physical-core-only affinity is applied. Setting to `32` disables affinity changes on most CPUs.
  - `AMDUseFirstCCDOnly` (default `true`): Activates first-CCD affinity on dual-CCD AMD CPUs (`CCD+` shown in tooltip).
  - `MSFSProcessPriority` (default `High`): Target MSFS priority level (`Normal` , `AboveNormal` , `High`).
  - `PowerPlanEnabled` (default `true`): Toggles automatic power-plan switching.

Simulator-Specific Features

- **MSFS 2020 Cloud Quality Reduction:** Drops volumetric clouds by one level if target FPS cannot be achieved at Base Min TLOD or if GPU load exceeds 98% (with GPU-Z running).
- **MSFS 2024 Auto Settings Reduction:** Progressively steps down non-LOD settings (shadows, trees, displacement maps, ray tracing) during sustained performance deficits when TLOD is already at its minimum, restoring them once headroom recovers.
- **MSFS 2024 Auto Increase Clouds:** Boosts cloud quality by one level if TLOD Base Extra is active and substantial performance margin remains.
- **MSFS 2024 Hybrid Dynamic Settings:** Allows AutoFPS to manage baseline LOD ranges while letting MSFS's internal dynamic settings engine handle micro-adjustments using native internal metrics.

Companion Apps & Advanced Features

GPU-Z Companion App & VRAM+ Protection

MSFS 2024 and MSFS 2020 in DX12 mode are very demanding on VRAM. When VRAM overflows into much slower system RAM, performance can drop to single digit FPS and MSFS becomes more prone to CTDs. VRAM+ monitors usage via GPU-Z shared memory and proactively holds or sheds settings to prevent the overflow.

Setup

1. Download and install GPU-Z.
2. In GPU-Z settings: enable *Minimise on close*, *Load at Windows startup (minimised)*, *Minimise to system tray*, and *Refresh sensors while in background*.
3. Make sure the card selected in the GPU-Z GPU dropdown is the GPU driving MSFS. GPU-Z shares the sensors of its selected card and AutoFPS cannot choose a different one.
4. VRAM+ is on by default, including in Non-Expert mode, and only works while GPU-Z is running.

VRAM+ Option (Expert mode)

- **MSFS 2020:** Enabling uses the LODs-only setting, providing up to a 50% reduction.
- **MSFS 2024: Off** disables the feature. **Max** (default) uses the full settings reduction suite minus clouds. **Set** uses the shared auto settings reduction options, whether or not auto settings reduction is enabled.
- Hover over the VRAM+ dropdown for a tooltip built from your live hold, reduce and recovery thresholds.

How it behaves

- **Hold** (default 96% VRAM in use): TLOD is capped at its current value, even if favourable performance conditions exist. The limit symbol (`T`) shows beside TLOD in Sim Values, with a tooltip.
- **Reduce** (default 98%): settings are progressively reduced (LODs by up to 50%, other settings down by up to 2 levels) until VRAM falls back below the Hold threshold. The **Reduce** label changes to **VRAM+** in red, and the status line appends **HLD** or **RED**.
- **Recovery** (default 93%, at least 5% below Reduce): settings progressively recover once VRAM has enough headroom, potentially back to their defaults.
- **Start Max:** is skipped when VRAM at session start is already above the recovery threshold, and cancelled if VRAM reaches the reduce threshold during the FPS settle. VRAM is monitored throughout the settle.
- If VRAM+ keeps activating, reduce your TLOD settings and/or other MSFS graphics settings.

Automation Thresholds

The three thresholds are changed in `MSFS2020_AutoFPS.config / MSFS2024_AutoFPS.config` (in `%appdata%\MSFS_AutoFPS`) after running the app once following an update:

- `VRAMOverflowHoldTLOD` (default `96` %): caps TLOD to its current value, preventing further upward seeking.
- `VRAMOverflowReduceTLOD` (default `98` %): progressively invokes settings reduction (or reduces LODs by up to 50%) until memory pressure stabilises.
- `VRAMOverflowSettingsRecovery` (default `93` %): must be at least 5% below the reduction threshold before settings are permitted to recover.

See *VRAM+ in full* in the Detailed Reference for every behaviour.

RTSS Companion App & Periodic Spike Protection

RivaTuner Statistics Server (RTSS) provides external frame timing metrics and enables detection of cyclic frame time stutter often encountered over dense photogrammetry.

- Spike Protection requires RTSS to be installed and running.
- It detects periodic frame time spikes (cadence of 0.3–1.8 seconds) matching photogrammetry reload stutters.
- When protection is active, AutoFPS applies temporary TLOD reductions, smoothly restoring values over time and distance once the spike train subsides.
- Logging is streamlined to record events only when 4 or more spikes occur outside a 1.8-second cadence window, minimising log file growth.
- `SpikeThresholdMultiplier` in `MSFS_AutoFPS.config` (default `2.0`) controls the multiple of the frame time median at which a spike is counted.

Monitor Only

Choose **Monitor only** at the end of the Flight Type list (Expert or Non-Expert mode) to run AutoFPS purely as a performance monitor and logger:

- AutoFPS stops changing any sim settings (the status line shows Monitor Mode). On selecting it, the sim is first returned to your own default TLOD, OLOD, cloud quality and key settings.
- It is remembered between sessions until you choose another flight type, which reloads that profile and does a full reset. It is not a profile, and your profiles are untouched.
- The layout is simplified across the desktop app, widget and web interface. Log+ is forced on and locked while you are in Monitor only.
- You can change TLOD, OLOD and cloud quality in the MSFS options at any time, in or out of a flight. The readings update immediately once you save, and nothing is restored or reapplied.
- VR, sim rate and Night profile auto-switching are suspended.

Log+ Diagnostic Mode & Performance Logging

Enable **Log+** in Expert mode to access advanced diagnostics:

- **Detailed FPS Logging:** Click the live FPS readout during flight to record raw and averaged FPS to the log file every second for a defined duration (indicated by alternating purple text). Default duration is 30 seconds, changeable via `logFPSDetailsCountMax` in `MSFS_AutoFPS.config`.
- **Camera State Logging:** Logs simulator camera transitions to help diagnose view-change stutters.
- **System Metrics Logging:** Logs total CPU load, dominant core usage, MSFS process usage, GPU load, and VRAM pressure using exponentially weighted moving averages (EWMA).
- **Config File Parameters (`MSFS_AutoFPS.config`):**
  - `perfLogTopCoreCount` (default `2`): Number of busiest cores tracked.
  - `PerfLogBufferSize` (default `15`): Pre-breach and post-recovery sample buffer.
  - `PerfLoggingAlways` (default `false`): Forces continuous logging when Log+ is active.
  - `EwmaAlpha` (default `0.1`): Smoothing coefficient for scheduler jitter.
  - `MonitorMode`: no longer enables monitoring. It only stores the Monitor only Flight Type selection, so choose **Monitor only** in the Flight Type list instead (see Monitor Only).

Troubleshooting & FAQ

Start here

- Installer will not download or run, or is flagged as unsafe: see Installer Problems.
- SimConnect stays red, TLOD and OLOD show zero, or a WASM exception is in the log: see SimConnect stays red, values show zero or a WASM exception.
- Need a clean MSFS reinstall: see MSFS Clean Reinstallation Instructions.
- "Unable to attach MSFS" or "MSFS compatibility test failed": see App Messages.
- Stutters, changing settings or VRAM+ questions: see the FAQ.

SimConnect stays red, values show zero or a WASM exception

This covers SimConnect staying red, TLOD and OLOD showing zero, "Critical Exception occurred: MSFS_AutoFPS - Unable to load DLL 'GpuzShMem.x64.dll' or one of its dependencies" in the log file, and a SimConnect / WASM exception such as Exception 31 or Exception 25. Work through these in order and stop as soon as it works:

1. **Repair.** Close MSFS, run the **MSFS_AutoFPS installer** (the one downloaded from the Releases page, not an MSFS installer), choose **Repair** and reboot. If a redistributable fails to install, install or repair these yourself:
   - Microsoft Visual C++ 2015 - 2022 Redistributable: x86 and x64.
   - The .NET Desktop Runtime x64 from the Microsoft .NET 8.0 download page.
2. **Remove duplicate MobiFlight modules.** Duplicates in both your Official and Community folders cause Sim Values to read zero. Remove the older copies, rerun the installer and start the app again.
3. **Delete the MSFS WASM folder** (when the log shows an exception such as 31 or 25, the WASM installation is most likely corrupt). It is located under the Microsoft Flight Simulator directory in either `%appdata%` (Steam) or `%localappdata%` (MS Store) and MSFS rebuilds it next time it runs. Rebooting is also recommended.
4. **Clean reinstall MSFS.** If that does not fix it, a full clean reinstall of MSFS is required: see MSFS Clean Reinstallation Instructions.

MSFS Clean Reinstallation Instructions

A clean reinstallation takes under 15 minutes in MSFS 2024, but several hours in MSFS 2020. Only undertake this if troubleshooting persistent WASM or core simulator crashes.

1. Back up or relocate your Community folder.
2. Follow the official Zendesk clean install instructions exactly: https://flightsimulator.zendesk.com/hc/en-us/articles/17335196046108-How-to-clean-install-the-simulator-on-PC
3. **Crucial:** Manually delete leftover simulator AppData/LocalState directories as specified in the guide. Controller profiles, logbooks, and cloud saves are retained on Xbox Live servers.
4. Reinstall the simulator, launch once to the main menu, and then restore your Community folder.

Installer Problems

- **Installer did not download or is zero size:** It should be about 3 MB. Try another browser, or temporarily disable your AV.

- **Installer will not run, or is flagged as untrusted:** Windows SmartScreen may block it because the app is new. Right-click the installer, choose Properties, tick **Unblock**, then Apply and OK, and try again. Otherwise create an exception for the installer in your AV program, or disable the AV temporarily while running it.
- **Cannot run the installer at all:** Follow the manual installation steps below.
- **Previously removed MSFS2020_AutoFPS without using its installer:** You may have problems installing MSFS_AutoFPS. Run the MSFS_AutoFPS installer and select remove, install MSFS2020_AutoFPS 0.4.3.1 with no autostart options, then rerun the MSFS_AutoFPS installer and choose your autostart option.
- **Bypass MobiFlight check:** If the installer fails during the MobiFlight check despite the module being current, create a shortcut to the installer, append `-bypassmobiflight` to the Target field, and run the shortcut.
- **Unable to locate Community folder:** Often caused by leftover simulator userdata following an MSFS uninstall. AutoFPS checks for `UserCfg.opt` to verify simulator presence, so removing orphan userdata folders resolves the false detection. Alternatively use the `2020 Folder...` or `2024 Folder...` button in the installer.
- **XML Exception during autostart:** "XML Exception: Unexpected XML declaration" indicates corruption in your `EXE.xml` file. Paste the file content into an XML validator or AI tool to correct the structure, save, and rerun the installer.
- **Installer error:** An unexpected installer error now shows the reason and next steps instead of leaving the installer stuck.

Manual Installation (Without Installer)

If the installer cannot be run:

1. Download the MobiFlight WASM Module and extract it into your MSFS Community folder (without MSFS running).
2. Install the Visual C++ Redistributable (x86 and x64).
3. Download and install the latest .NET 8 Desktop Runtime (x64).
4. Create the folder `%appdata%\MSFS_AutoFPS\bin`.
5. Download MSFS_AutoFPS.zip and extract its contents into the newly created `bin` folder.
6. Create a shortcut to `MSFS_AutoFPS.exe` on your desktop.

App Messages

- **"Unable to attach MSFS - app disabled":** The most likely causes are MSFS loading very slowly so the attachment times out, MSFS and AutoFPS running at different permission

levels (e.g. MSFS running as Admin while AutoFPS is not), or your anti-virus blocking the app. Restart AutoFPS after MSFS has loaded to the main menu, check that the elevation levels match, set an anti-virus exclusion for the app, and as a last resort run AutoFPS as administrator.

- **"MSFS compatibility test failed - app disabled":**
  - You started MSFS, changed MSFS settings and then started AutoFPS. Exit AutoFPS, toggle any simple setting (e.g. VSync) in the MSFS settings menu, save, and restart AutoFPS. If that does not work, exit both completely, start AutoFPS, then start MSFS.
  - `UserCfg.opt` contains values beyond what the MSFS menu allows. Reset the graphics settings to default (F12) in the MSFS main menu settings for both PC (MSFS 2020) or General (MSFS 2024) and VR, then make all changes within the MSFS settings menu.
  - A new MSFS version may have moved the memory map. AutoFPS will first auto-install an explicitly compatible update if one is available, otherwise it offers the update most likely to be compatible, which may be a test build. If none is found or it does not help, I am likely already aware and working on it. If you may be among the first to hit it (for example on an MSFS beta), please raise a new issue on GitHub or add to an existing one.
  - There may be a permissions issue and you may need to run the app as Administrator.

Frequently Asked Questions

How does AutoFPS compare to the in-built Dynamic Setting feature in MSFS 2024?

- **Independent Adjustment:** MSFS Dynamic Settings adjusts TLOD first and then OLOD if performance remains poor. AutoFPS manages TLOD and OLOD independently, with OLOD adjusting inversely proportional to altitude by default. AutoFPS can also manage a wider suite of visual settings when Settings Reduction and VRAM+ are active.
- **Adjustment Range:** MSFS Dynamic Settings only reduces down to 50% of your configured setting. AutoFPS in Non-Expert mode uses defaults as a midpoint, and in Expert mode can be set to any range, including beyond the in-game menu maximum of 400.
- **Response Speed:** MSFS Dynamic Settings responds faster due to internal metric access, but tends to overreact to brief stutters. AutoFPS filters short-term spikes to maintain visual stability.
- **Frame Generation Awareness:** MSFS Dynamic Settings bases its target on native framerate rather than delivered framerate when frame generation is active. AutoFPS accommodates all mainstream frame generation technologies with post-FG targets.

- **Hybrid Automation:** In Expert Mode, AutoFPS can work alongside MSFS Dynamic Settings: MSFS handles rapid micro-adjustments while AutoFPS manages base LOD ranges and external settings reductions.

Why does my antivirus or browser flag the installer as untrusted?

AutoFPS is a hobbyist project. Code signing certificates carry prohibitive costs, so new releases may occasionally trigger false positives (Windows SmartScreen, Kaspersky, or browser download warnings) until community reputation builds. If blocked, right-click the installer, select **Properties**, check **Unblock**, and click **Apply**. Downloaded updates are also verified by AutoFPS against the SHA-256 digest published by GitHub before they are launched.

Why are my simulator settings restoring to unexpected values?

AutoFPS records your default settings when a flight begins and restores them when returning to the main menu or exiting the app. If you alter graphics settings in the MSFS menu while in an active flight with AutoFPS running, those changes will be overwritten. To change baseline settings, do so from the MSFS main menu or with AutoFPS closed, or choose **Monitor only**, where AutoFPS does not change or restore anything.

Can you tell me the best settings for me to use?

- The best settings depend on your system, the other addons and apps you use, and what you mean by "best": higher FPS, better visuals or a smoother experience. Some of these are mutually exclusive.
- There are now thousands of users of this app and I don't have the time to give individual settings advice.
- Start with Non-Expert mode to see what the app does and when. When you are comfortable, enable Expert and read the tooltip and README for each setting.
- Make small changes to only a few settings at a time so you can see the effect of each.

How do I eliminate visual or audio stutters when using the app?

1. Revert to the default configuration using the installer's **Reset Configs** option.
2. Begin in Non-Expert mode with the IFR profile and Auto Target FPS.
3. If the MSFS Performance Optimiser is enabled, try disabling it, as core affinity changes can cause audio crackling on certain hardware.
4. In Expert mode, lower **FPS Sensitivity** to 2 or 3 to enforce smaller, gradual TLOD adjustments.

What does VRAM+ do and why does it show the wrong GPU or not work?

VRAM+ holds or reduces key settings when VRAM is about to overflow (see GPU-Z Companion App & VRAM+ Protection). It only works while GPU-Z is running. GPU-Z shares the sensors of the card selected in its own GPU dropdown, and AutoFPS cannot choose a different card, so on a system with more than one GPU select the card that drives MSFS in GPU-Z.

Why does RTSS framerate telemetry freeze?

Switching between 2D Frame Generation and VR can latch the swapchain in shared memory. Clicking the FPS source icon on the app or widget toggles capture back to MSFS. If RTSS remains frozen, restarting RTSS or MSFS clears the shared memory handle.

Detailed Reference

The sections below hold the full technical detail from the original README. Click a heading to expand it.

▶ **Credits and licence history**

▶ **Graphics modes: full behaviour of each mode**

▶ **Connection status, Sim Values, FPS source, updates, VRAM+, hybrid dynamic settings and status messages**

▶ **VRAM+ in full**

▶ **Target FPS options, profiles, Non-Expert defaults and Expert options overview**

▶ **Expert settings in full (automation methods, TLOD, Night Mode, Auto OLOD, cloud quality, settings reduction, VRAM+)**

▶ **Log+ mode and performance logging**

▶ **General operation and app window notes**

▶ **Installation, update and uninstall notes in full**

▶ **FAQ answers in full**